

Project Design Phase – II

Data Flow Diagram & User Stories

Date	31 January 2025
Team ID	HYD-METRO-2025
Project Name	Hyderabad Metro Rail – Digital Ticketing System (ServiceNow)
Maximum Marks	4 Marks

Data Flow Diagrams (DFD)

A Data Flow Diagram (DFD) is a traditional visual representation of the information flows within a system. It shows how data enters and leaves the system, what changes the information, and where data is stored. Below are the DFD Level 0 and Level 1 for the Hyderabad Metro Rail Digital Ticketing System.

DFD Level 0 – Context Diagram (System Overview)

The Level 0 DFD shows the system as a single process and its interaction with external entities.

External Entity	Data Flow (Input to System)	Process	Data Flow (Output from System)	External Entity
Commuter (End User)	Booking Request (Source, Destination, Passengers, Journey Type, Payment Mode)	Metro Digital Ticketing System (ServiceNow)	QR Code Ticket + Booking Confirmation + Fare Amount	Commuter (End User)
Station Manager	Login Credentials + Access Request	Metro Digital Ticketing System (ServiceNow)	Booking Reports + Analytics Dashboard	Station Manager
IT Administrator	System Config + ACL Rules + Flow Activation	Metro Digital Ticketing System (ServiceNow)	System Status + Audit Logs	IT Administrator
Payment Gateway (Future)	Payment Confirmation + Transaction ID	Metro Digital Ticketing System (ServiceNow)	Payment Request + Amount	Payment Gateway (Future)

DFD Level 1 – Detailed Process Breakdown

The Level 1 DFD breaks down the core system process into sub-processes and data stores.

Process #	Process Name	Input Data	Output Data	Data Store
P1	User Authentication & Portal Access	Username + Password	Session Token / Portal Access	User Table (sys_user)
P2	Catalog Item Display & Selection	Portal Search Query	Book A Metro Ticket Form	Service Catalog (sc_cat_item)
P3	Form Variable Capture	Source, Destination, No. of Passengers, Journey Type	Payment Request + Amount	Catalog Variables (sc_item_option)
P4	Fare Auto-Calculation (Client Script)	Source sys_id + Destination sys_id	Fare Amount (Single / Return / Return + Stop)	Station Details Table

P5	Form Validation (Client Script)	No. of Passengers field input	Validation Alert / Pass	None (client-side)
P6	QR Code Generation (OnSubmit Script)	Submitted Form + Requested Item Sys_id	QR Code URL via GlideModal	None (client-side display)
P7	Flow Designer Trigger (Metro Project)	Requested Item (RITM)	Flow Execution Signal	Flow Designer Engine
P8	Get Catalog Variables (Flow Action 1)	RITM sys_id	Variable Values (all fields)	sc_item_option table
P9	Create Record (Flow Action 2)	Mapped Variable Values	New Booking Record	Metro Database (custom table)
P10	ACL Enforcement	User Role + Table Access Request	Access Granted / Denied	ACL Rules (sys_security_acl)
P11	Email Notification	Booking Record + User Email	Booking Confirmation Email	Notification Engine

User Stories

User stories define the functionality of the system from the perspective of each user type. Each story follows the format: As a [user], I can [action] so that [benefit].

User Type	Epic (FR)	USN	User Story / Task	Acceptance Criteria	Priority	Release
Commuter (Portal User)	Ticket Booking	USN-001	As a commuter, I can open the ServiceNow portal to view the Metro Catalog.	Catalog from Books & More Searchable & Filterable	High	Spring 2024
Commuter (Portal User)	Ticket Booking	USN-002	As a commuter, I can select my starting station and destination from a dropdown list of metro stations.	Reference dropdown for all metro stations	High	Spring 2024
Commuter (Portal User)	Ticket Booking	USN-003	As a commuter, I can enter the number of passengers, and the system will validate; alert shown for invalid input.	Field accepts only numeric values; alert shown for invalid input.	High	Spring 2024
Commuter (Portal User)	Fare Calculation	USN-004	As a commuter, I can see the fare auto-populated based on my route and journey type. No manual entry needed.	Fare calculated automatically based on route and journey type	High	Spring 2024
Commuter (Portal User)	Journey Type	USN-005	As a commuter, I can select Single or Return journey. The system will calculate the correct fare based on the selection.	Single fare 1x base; Return fare 2x base. Correct field populated.	High	Spring 2024
Commuter (Portal User)	Payment	USN-006	As a commuter, I can select my payment mode (UPI, Card, Wallet, Cash, etc.). After payment, the 'Payment Mode' field appears.	UPI, Card, Wallet, Cash, etc. Payment Mode field appears	High	Spring 2024
Commuter (Portal User)	QR Code	USN-007	As a commuter, upon submitting the booking form, the system will generate a QR code for the journey. The QR code can be scanned with a scannable app.	QR code generated for the booking. Scannable with a scannable app	High	Spring 2024
Commuter (Portal User)	Ticket Booking	USN-008	As a commuter, I can submit the form and get a booking confirmation. The confirmation is sent to the user's email and is visible in the 'My Requests' section.	Booking generated, confirmation sent via email and shown in 'My Requests'	High	Spring 2024
Commuter (Portal User)	Metro Card	USN-009	As a commuter, I can choose 'Recharge Metro Card' from the catalog. The system will show the recharge amount and options.	Recharge option in catalog. Amount and options shown	Medium	Spring 2024
Commuter (Portal User)	Notifications	USN-010	As a commuter, I receive an email confirmation after my booking is successfully submitted. The email is sent within 5 minutes of submission.	Email is sent after successful booking. Within 5 minutes	High	Spring 2024
Station Manager	Data Access	USN-011	As a station manager, I can view all booking records for a specific station. The data is visible in the Metro Database.	Table in the Metro Database; records show all mapped fields	High	Spring 2024
Station Manager	Reporting	USN-012	As a station manager, I can filter bookings by date, route, and passenger count. The system will display the filtered data in a table.	Date, route, and passenger count filters. Data displayed in table	Medium	Spring 2024
Station Manager	Analytics	USN-013	As a station manager, I can view a dashboard showing booking trends, peak times, and passenger counts. The dashboard will provide insights into booking patterns.	Dashboard with graphs for booking trends, peak times, and passenger counts	Medium	Spring 2024
IT Admin	Access Control	USN-014	As an IT admin, I can configure ACL rules so that only authorized users can access the Metro Database. The system will enforce these rules.	ACL rules configured to restrict access to Metro Database	High	Spring 2024
IT Admin	Flow Config	USN-015	As an IT admin, I can activate and monitor the Metro Projective Flow. The system will log any errors and provide alerts.	Projective Flow activated. Error logs and alerts configured	High	Spring 2024
IT Admin	Maintenance	USN-016	As an IT admin, I can add or update metro station details in the database. The system will ensure data consistency and provide backup/restore options.	Station details added/updated. Data consistency maintained. Backup/restore options available	High	Spring 2024